

HOMEWORK 4

INTRODUCTION TO COMPLEX ANALYSIS

(MATH 4230-40), SPRING 2022

SECTION 1.6 - DIFFERENTIATION OF THE ELEMENTARY FUNCTIONS

1. Determine whether the following complex limits exist and find their value if they do:

(a) $\lim_{z \rightarrow 0} \frac{e^z - 1}{z}$

Recall the definition of the derivative: $f'(a) := \lim_{z \rightarrow a} \frac{f(z) - f(a)}{z - a}$. So, letting $f(z) = e^z$ and observing that $f'(z) = e^z$ by Proposition 1.6.1, note that we have

$$\begin{aligned} \lim_{z \rightarrow 0} \frac{e^z - 1}{z} &= \lim_{z \rightarrow 0} \frac{e^z - e^0}{z - 0} \\ &= \lim_{z \rightarrow 0} \frac{f(z) - f(0)}{z - 0} \\ &= f'(0) \\ &= e^0 \\ &= 1. \end{aligned}$$

(b) $\lim_{z \rightarrow 0} \frac{\sin |z|}{z}$

To show that this limit does not exist, it is enough to show that approaching 0 from two different directions yields two different values. Suppose that z approaches 0 along the positive real axis; e.g. as $z = x + iy \rightarrow 0$ when $y = 0$ and $x > 0$. This produces

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{\sin x}{x} &= \lim_{x \rightarrow 0^+} \frac{\sin x - \sin 0}{x - 0} \\ &= \cos(0) \\ &= 1. \end{aligned}$$

Similarly, we could suppose that z approaches 0 along the positive imaginary axis; e.g. as $z = x + iy \rightarrow 0$ when $x = 0$ and $y > 0$. This produces

$$\begin{aligned} \lim_{y \rightarrow 0^+} \frac{\sin y}{iy} &= \frac{1}{i} \lim_{y \rightarrow 0^+} \frac{\sin y - \sin 0}{y - 0} \\ &= \frac{1}{i} \cos(0) \\ &= \frac{1}{i} \\ &= -i. \end{aligned}$$

As $1 \neq -i$, the limit does not exist.

2. Let $u(x, y)$ and $v(x, y)$ be real valued functions on an open set $A \subseteq \mathbb{R}^2$ and suppose they satisfy the Cauchy-Riemann equations on A . Show that this implies that the following pairs of functions satisfy the Cauchy-Riemann equations on A as well:

(a) $u_1 := (u(x, y))^2 - (v(x, y))^2$ and $v_1(x, y) := 2u(x, y)v(x, y)$

Suppose that $u(x, y)$ and $v(x, y)$ are real valued functions on an open set $A \subseteq \mathbb{R}^2$ that satisfy the Cauchy-Riemann equations on A . We seek to show that u_1 and v_1 satisfy the Cauchy-Riemann equations on A as well. To do so, we first verify that $\frac{\partial v_1}{\partial y} = \frac{\partial u_1}{\partial x}$:

$$\begin{aligned}\frac{\partial v_1}{\partial y} &= 2 \frac{\partial u}{\partial y} v(x, y) + 2 \frac{\partial v}{\partial y} u(x, y) \quad (\text{Product rule}) \\ &= -2 \frac{\partial v}{\partial x} v(x, y) + 2 \frac{\partial v}{\partial y} u(x, y) \quad \left(\text{Since } \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \right) \\ &= 2u(x, y) \frac{\partial u}{\partial x} - 2v(x, y) \frac{\partial v}{\partial x} \quad \left(\text{Since } \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \right) \\ &= \frac{\partial u_1}{\partial x} \quad (\text{Chain rule})\end{aligned}$$

Now, we verify that $\frac{\partial u_1}{\partial y} = -\frac{\partial v_1}{\partial x}$:

$$\begin{aligned}\frac{\partial v_1}{\partial x} &= 2 \frac{\partial u}{\partial x} v(x, y) + 2 \frac{\partial v}{\partial x} u(x, y) \quad (\text{Product rule}) \\ &= 2 \frac{\partial u}{\partial x} v(x, y) - 2 \frac{\partial u}{\partial y} u(x, y) \quad \left(\text{Since } \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \right) \\ &= 2 \frac{\partial v}{\partial y} v(x, y) - 2 \frac{\partial u}{\partial y} u(x, y) \quad \left(\text{Since } \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \right) \\ &= -\left(2 \frac{\partial u}{\partial y} u(x, y) - 2 \frac{\partial v}{\partial y} v(x, y) \right) \quad (\text{Factoring out } -1) \\ &= -\frac{\partial u_1}{\partial y} \quad (\text{Chain rule})\end{aligned}$$

Therefore, we have shown that u_1 and v_1 satisfy the Cauchy-Riemann equations on A as well.

(b) $u_2(x, y) := e^{u(x, y)} \cos(v(x, y))$ and $v_2(x, y) := e^{u(x, y)} \sin(v(x, y))$

Suppose that $u(x, y)$ and $v(x, y)$ are real valued functions on an open set $A \subseteq \mathbb{R}^2$ that satisfy the Cauchy-Riemann equations on A . We seek to show that u_2 and v_2 satisfy the Cauchy-Riemann equations on A as well. To do so, we first verify that $\frac{\partial v_2}{\partial y} = \frac{\partial u_2}{\partial x}$:

$$\begin{aligned}
 \frac{\partial u_2}{\partial x} &= \left(\frac{\partial}{\partial x} e^{u(x, y)} \right) \cos(v(x, y)) + e^{u(x, y)} \left(\frac{\partial}{\partial x} \cos(v(x, y)) \right) \quad (\text{Product rule}) \\
 &= \frac{\partial u}{\partial x} e^{u(x, y)} \cos(v(x, y)) - e^{u(x, y)} \frac{\partial v}{\partial x} \sin(v(x, y)) \quad (\text{Chain rule}) \\
 &= \frac{\partial v}{\partial y} e^{u(x, y)} \cos(v(x, y)) - e^{u(x, y)} \frac{\partial v}{\partial x} \sin(v(x, y)) \quad \left(\text{Since } \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \right) \\
 &= \frac{\partial v}{\partial y} e^{u(x, y)} \cos(v(x, y)) + e^{u(x, y)} \frac{\partial u}{\partial y} \sin(v(x, y)) \quad \left(\text{Since } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \right) \\
 &= e^{u(x, y)} \left(\frac{\partial}{\partial y} \sin(v(x, y)) \right) + \left(\frac{\partial}{\partial y} e^{u(x, y)} \right) \sin(v(x, y)) \quad (\text{Chain rule}) \\
 &= \frac{\partial v_2}{\partial y} \quad (\text{Product rule})
 \end{aligned}$$

Now, we verify that $\frac{\partial u_2}{\partial y} = -\frac{\partial v_2}{\partial x}$:

$$\begin{aligned}
 \frac{\partial u_2}{\partial y} &= \left(\frac{\partial}{\partial y} e^{u(x, y)} \right) \cos(v(x, y)) + e^{u(x, y)} \left(\frac{\partial}{\partial y} \cos(v(x, y)) \right) \quad (\text{Product rule}) \\
 &= \frac{\partial u}{\partial y} e^{u(x, y)} \cos(v(x, y)) - e^{u(x, y)} \frac{\partial v}{\partial y} \sin(v(x, y)) \quad (\text{Chain rule}) \\
 &= \frac{\partial u}{\partial y} e^{u(x, y)} \cos(v(x, y)) - e^{u(x, y)} \frac{\partial u}{\partial x} \sin(v(x, y)) \quad \left(\text{Since } \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \right) \\
 &= -\frac{\partial v}{\partial x} e^{u(x, y)} \cos(v(x, y)) - e^{u(x, y)} \frac{\partial u}{\partial x} \sin(v(x, y)) \quad \left(\text{Since } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \right) \\
 &= -\left(\frac{\partial v}{\partial x} e^{u(x, y)} \cos(v(x, y)) + e^{u(x, y)} \frac{\partial u}{\partial x} \sin(v(x, y)) \right) \quad (\text{Factoring out } -1) \\
 &= -\left(e^{u(x, y)} \left(\frac{\partial}{\partial x} \sin(v(x, y)) \right) + \left(\frac{\partial}{\partial x} e^{u(x, y)} \right) \sin(v(x, y)) \right) \quad (\text{Chain rule}) \\
 &= -\frac{\partial v_2}{\partial x} \quad (\text{Product rule})
 \end{aligned}$$

Therefore, we have shown that u_2 and v_2 satisfy the Cauchy-Riemann equations on A as well.

3. Find the region of analyticity and the derivative of each of the following functions:

(a) $\frac{z}{z^2 - 1}$

By Proposition 1.5.3 part (v), any rational function is analytic on the open set consisting of all $z \in \mathbb{C}$ except those points where the denominator is zero. As the function $f(z) = \frac{z}{z^2 - 1}$ is rational and $z^2 - 1 = 0$ when $z = \pm 1$, it follows that f is analytic on the set $\mathbb{C} \setminus \{-1, 1\}$. By Proposition 1.5.3 part (iv) and Proposition 1.5.3 part (iii), the derivative of f is given by

$$f'(z) = \frac{(1)(z^2 - 1) - (2z)(z)}{(z^2 - 1)^2} = \frac{-1 - z^2}{(z^2 - 1)^2}.$$

(b) $e^{z+(1/z)}$

Note that $f(z) = e^{z+1/z}$ can be re-written as $f(z) = e^{\frac{z^2+1}{z}}$. As e^z is entire by Proposition 1.6.1 and $\frac{z^2+1}{z}$ is analytic on $\mathbb{C} \setminus \{0\}$ by Proposition 1.5.3 part (v), the chain rule (i.e., Theorem 1.5.4) yields that $f(z)$ is analytic on $\mathbb{C} \setminus \{0\}$. Additionally, employing Proposition 1.6.1, Proposition 1.5.3 part (iv), and Proposition 1.5.3 part (iii) in the context of the chain rule, we obtain

$$f'(z) = \frac{(2z)(z) - (z^2 + 1)(1)}{z^2} e^{\frac{z^2+1}{z}} = \frac{z^2 - 1}{z^2} e^{z+(1/z)}.$$